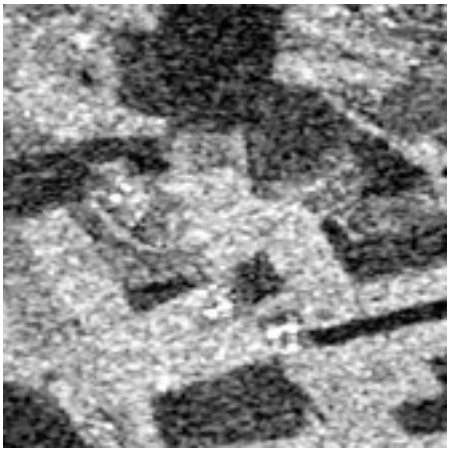
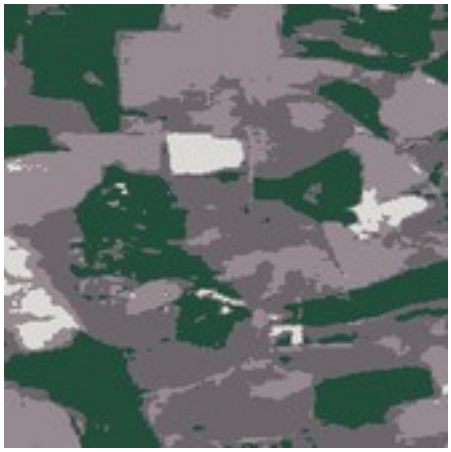
SAR Image Analysis Report



Original SAR

Colorized



K-Means Segmented

Metadata:

- Original File: original_1764435982.png
- Colorized File: colorized_1764435982.png
- Processed On: 2025-11-29 22:36:42 IST
- SAR Classifier Confidence: 0.981
- Colorization Model: Pix2Pix GAN
- KMeans Clusters: 4
- Colorized Resolution: 256 x 256 px

Cluster Analysis:

Cluster 0: Likely Vegetation (Green Tone) | Mean RGB: (35, 78, 60) | Area: 31.01%

Cluster 1: Likely Urban / Built-up (Bright/Gray) | Mean RGB: (150, 139, 144) | Area: 27.37%

Cluster 2: Likely Soil / Barren Land (Reddish Tone) | Mean RGB: (109, 102, 107) | Area: 37.64%

Cluster 3: Likely Urban / Built-up (Bright/Gray) | Mean RGB: (221, 219, 217) | Area: 3.98%

Land Cover Summary:

- Likely Vegetation (Green Tone) → 31.01%
- Likely Urban / Built-up (Bright/Gray) → 27.37%
- Likely Soil / Barren Land (Reddish Tone) → 37.64%
- Likely Urban / Built-up (Bright/Gray) → 3.98%

Note: Colorized image is GAN-generated and for visualization only.

Analysis Details & Recommendations

Interpretation Notes:

The area is dominated by likely soil / barren land (reddish tone) (37.64% combined).

Secondary presence: likely urban / built-up (bright/gray) (31.35%).

Secondary presence: likely vegetation (green tone) (31.01%).

K-Means groups pixels by color similarity — labels are heuristic.

Consider adding multispectral or time-series data for improved analysis.

Recommendations:

The analysis shows dominant land cover is soil/barren (37.07%). Large soil/barren area suggests land degradation or exposed surfaces; consider erosion control or land restoration planning. The second-largest class is vegetation (30.82%), so combined management strategies addressing both soil/barren and vegetation are advisable. Recommended actions: collect ground-truth samples where possible, prioritize monitoring based on dominant class, and re-run analysis after incorporating higher-resolution data or additional bands.

Confidence Indicators:

SAR Detection Confidence: 0.981
Dominant Class Percentage: 37.64%
Classification Reliability: High

Limitations:

- KMeans is unsupervised and may not match exact land-cover classes.
- GAN colorization is synthetic and does not represent true spectral values.
- Classification depends on image quality and may contain uncertainty.
- Single-image analysis; use multi-temporal data for trend-based conclusions.

Final Summary:

The analysis indicates predominant vegetation with mixed surfaces.

No major water bodies or dense urban zones detected in this image.

Recommend periodic monitoring and ground-truth sampling to improve confidence.